HW 5.1: In this question we will explore one last aspect of internal corrections to the Hydrogen atom (hyperfine splitting, HFS), as well as the response of the atom to an external magnetic field (the Zeeman effects). We work in natural units, so $\hbar = c = 1$ and $\alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137}$ in what follows.

Previously, we considered the total angular momentum and its z projection (j, m_j) as good quantum numbers for the Hydrogen atom and its fine structure perturbations. There, the spin-orbit coupling dependend on $\mathbf{L}\cdot\mathbf{S}$, which commuted with L^2, S^2, and J^2, but not with L_z and S_z.

In this first problem, we will consider the hyperfine structure of the Hydrogen atom. We will tackle the case $\ell \neq 0$ where we will need to account for the magnetic field induced by the orbital motion of the electron or nucleus. (The $\ell = 0$ case is solved in Griffith's book, and he only needs to account for the interaction between spins!) Note that this is harder than our previous S.O. problem: before, we had a dipole (the electron with a spin) interacting with the magnetic field generated by a moving charge (the spinless proton). Now, we have two dipoles (the electron and the nucleus with their spins) interacting with each other, so we have a 1) spin-spin coupling, 2) the proton-spin-electron-orbit coupling, and finally 3) the old electron-spin-proton-orbit coupling we calculated. The latter is unchanged, but the first two are new and we will calculate them.

We will interpret the problem as the interaction of the nuclear spin $\mathbf{I}$ with two internal magnetic fields calculated at the location of the nucleus: 1) the B-field produced by the electron spin (a dipole), and 2) the B-field produced by the orbital motion of the electron:

$$H_{\text{hfs}} = -\mathbf{M}_I \cdot \mathbf{B}_{\text{el}} = -\mathbf{M}_I \cdot \left(\mathbf{B}^\ell_{\text{el}} + \mathbf{B}^s_{\text{el}}\right), \tag{1}$$

where $\mathbf{M}_I$ is the nuclear magnetic moment and $\mathbf{B}_{\text{el}}$ is the magnetic field produced by the electron at the location of the nucleus (sorry, my boldface μ does not work for some reason, so we will go with $\mathbf{M}_I$ instead). Similarly to what we argued in class, the magnetic field produced by the orbital motion of the electron is given by

$$\mathbf{B}^\ell_{\text{el}} = -\mathbf{v}\times\mathbf{E} = \frac{\mathbf{E}\times\mathbf{p}}{m_e} = -\frac{1}{Ze}\frac{dV(r)}{dr}\frac{\mathbf{r}\times\mathbf{p}}{m_e r} = -\frac{\alpha}{e m_e}\frac{\mathbf{L}}{r^3}, \tag{2}$$

where we used the fact that $\mathbf{L} = \mathbf{r}\times\mathbf{p}$, $V(r) = -\frac{Z\alpha}{r}$, and $\mathbf{F} = +Ze\mathbf{E} = -\nabla V(r) = -\frac{dV(r)}{dr}\hat{\mathbf{r}}$ (note this now has a positive charge and the radial coordinate is the same we are used to, namely, $\mathbf{r}$ points from the nucleus to the electron). The magnetic field produced by the electron spin is given by the standard formula for the magnetic field of a dipole:

$$\mathbf{B}^s_{\text{el}} = -\frac{g_e\alpha}{e m_e}\left[\frac{1}{r^3}\left(3(\mathbf{S}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{S}\right) - \frac{8\pi}{3}\mathbf{S}\delta^3(\mathbf{r})\right], \tag{3}$$

where the first term is a standard result in E&M (the field of a dipole looks like a quadrupole from far away) and the second accounts for what happens right at the location of the origin (if we think of spin as an infinitesimal current loop, then the magnetic field at the center of the loop blows up, hence the delta function term). This delta function term is called the Fermi contact term. The factor of g_e is the electron g-factor, which we said was $\simeq 2$.

Putting everything together, the HFS perturbation is given by[1]

$$H_{\text{hfs}} = -\mathbf{M}_I\cdot\mathbf{B}_{\text{el}} = \frac{\alpha g_I}{2m_N m_e}\mathbf{I}\cdot\mathbf{A} \text{ where } \boxed{\mathbf{A} \equiv \frac{\mathbf{L}}{r^3} + g_e\left(\frac{1}{r^3}\left(3(\hat{\mathbf{r}})(\mathbf{S}\cdot\hat{\mathbf{r}}) - \mathbf{S}\right) - \frac{8\pi}{3}\mathbf{S}\delta^3(\mathbf{r})\right).} \tag{4}$$

[1]Note the absence of the Thomas precession factor of 1/2 in this calculation since we have not changed frames; before it arose because the electron rest frame was not inertial.

We used the nuclear magnetic moment $\mathbf{M}_I = g_I \frac{e}{2m_N}\mathbf{I}$, where g_I is the nuclear g-factor and m_N is the nucleon mass. In what follows, you may assume that our nucleus has spin $I = 1/2$ (this is the case for the proton, but not for all nuclei).

Now, we define the total angular momentum of the system as

$$\mathbf{F} = \mathbf{I} + \mathbf{J}$$

where we expect the new good quantum numbers to be (n, l, j, I, f, m_f).

a) Which of the following operators does the HFS perturbation commute with: J^2, J_z, S^2, S_z, I^2, I_z, F^2, F_z?

b) Starting from the irreps of the three angular momentum operators $\mathbf{S}$, $\mathbf{L}$, and $\mathbf{I}$, construct the irreps of $\mathbf{F}$. In other words, decompose $\mathbf{2} \otimes \mathbf{2} \otimes (2\mathbf{l}+1)$ into direct sums of irreps of $\mathbf{F}$. What is the range of values that f can take for a given j and I? What about m_f? What are the $n = 1$ states (previously, the degenerate ground states)?
(Hint: recall that in the spin-orbit problem, we had to decompose $\mathbf{2} \otimes \mathbf{2}$ into irreps of $\mathbf{J}$, and we found that j could take values $|l - s|$ to $l + s$.)

c) Calculate the total HFS energy correction for the $\ell = 0$ states.
(Hint: for $\ell = 0$, use the fact that the wavefunctions are spherically symmetric and express $\mathbf{I} \cdot \mathbf{S}$ in terms of operators that commute with $\mathbf{F}$.)

d) Calculate the HFS energy correction for the $\ell \neq 0$ states.
(Hint: for $\ell \neq 0$, the wavefunctions are not spherically symmetric and we have not choice but to calculate the expectation value of some nasty operators. Thankfully, the Wigner-Eckart theorem comes to our aid to gives a very useful relation:

$$\mathbf{A} \longrightarrow \frac{(\mathbf{A} \cdot \mathbf{J})}{j(j+1)}\mathbf{J} \tag{5}$$

where we have used the fact that $\mathbf{J}$ are the only vectors that we can construct from $\mathbf{A}$ that have the right transformation properties under rotations, and we have fixed the proportionality constants by taking the dot product of both sides with $\mathbf{J}$.)

e) Express the HFS energy for the ground state of Hydrogen (the 1S state) in eV and then in Hz. What photon energy would correspond to this transition in centimeters? In what frequency band does this transition lie? Which of the following drawings on the voyager plate is alluding to this transition?

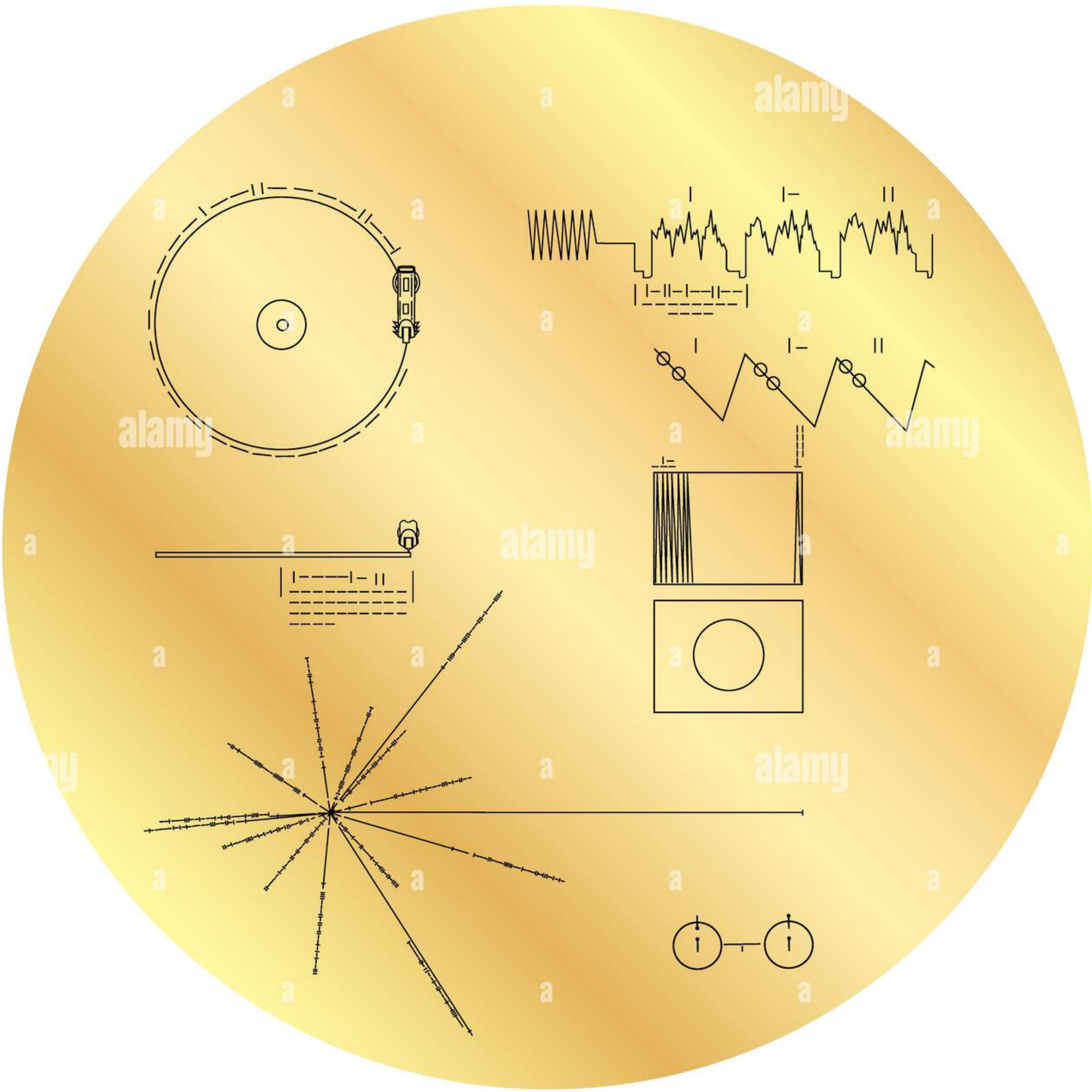

Figure 1: The cover of the Voyager Golden Record (reproduction of the original record sent to space in 1977).

HW 5.2: Consider now an external magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$. The Zeeman effect is the splitting of energy levels due to the interaction of the magnetic field with the magnetic moments of the electron and nucleus. For now, let us ignore the nuclear spin and focus on the interaction of the magnetic field with the electron.

Including the electron-spin-orbit coupling, the total Hamiltonian of the system is given by

$$H_{\text{Zeeman}} = H_0 + \frac{g_e \alpha}{4m_e^2 r^3}\mathbf{L}\cdot\mathbf{S} - \frac{e}{2m_e}(L_Z + g_e S_z)B. \tag{6}$$

where we will set $g_e = 2$.

a) Which of the following operators does the Zeeman perturbation commute with: J^2, J_z, S^2, S_z, L^2, L_z?

b) Calculate the correction to the energy levels due to the Zeeman effect in the weak field limit (where the Zeeman perturbation is smaller than the fine structure spin-orbital terms).
(Hint 1: first find the good quantum numbers in this case).
(Hint 2: again, use $S_z \longrightarrow \frac{(\mathbf{S}\cdot\mathbf{J})}{j(j+1)}J_Z$ and the fact that $j = l \pm \frac{1}{2}$, unless $\ell = 0$, when $j = \frac{1}{2}$.
)

c) Calculate the correction to the energy levels due to the Zeeman effect in the strong field limit, neglecting the spin-orbit term.
(Hint: Recall that angular momentum conservation is a consequence of spherical symmetry, so if we break it, we should not expect to have good quantum numbers associated with J^2, for example. Instead, L_z and S_z will be good quantum numbers since the system is still symmetric under rotations around the z-axis.)

d) What happens in the intermediate field limit? What would be your plan of attack to solve the problem in this case? (Hint: do not try to do it.)